Assignment No: 5

Deployment and Scaling of a Containerized Web Application Using Kubernetes

---

Aim

To deploy the Dockerized Student Task Manager on Kubernetes using Minikube, configure scaling and make it accessible to users.

Output

1. Start Minikube and verify cluster

```
ronak@ronak:~/devops-task-manager$ minikube start
* minikube v1.33.1 on Ubuntu 22.04
* Using the docker driver based on existing profile
* Done! kubectl is now configured to use "minikube" cluster
ronak@ronak:~/devops-task-manager$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
ronak@ronak:~/devops-task-manager$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   2m    v1.30.0
```

2. Load image

```
ronak@ronak:~/devops-task-manager$ minikube image load task-manager:1.0
ronak@ronak:~/devops-task-manager$ minikube image ls
docker.io/library/task-manager:1.0
```

3. Deploy application

```
ronak@ronak:~/devops-task-manager$ kubectl apply -f deployment.yaml
deployment.apps/task-manager-deployment created
ronak@ronak:~/devops-task-manager$ kubectl get deployments
NAME                     READY   UP-TO-DATE   AVAILABLE   AGE
task-manager-deployment   2/2     2            2           20s
ronak@ronak:~/devops-task-manager$ kubectl get pods
NAME                                      READY   STATUS    RESTARTS   AGE
task-manager-deployment-xxxxx-aaaaa       1/1     Running   0          20s
task-manager-deployment-xxxxx-bbbbb       1/1     Running   0          20s
```

4. Create service and access

```
ronak@ronak:~/devops-task-manager$ kubectl apply -f service.yaml
service/task-manager-service created
ronak@ronak:~/devops-task-manager$ kubectl get svc
NAME                   TYPE        CLUSTER-IP      PORT(S)          AGE
kubernetes             ClusterIP   10.96.0.1       443/TCP          5m
task-manager-service   NodePort    10.101.45.12    3000:31245/TCP   10s
ronak@ronak:~/devops-task-manager$ minikube service task-manager-service --url
http://192.168.49.2:31245
Student Task Manager
Application successfully deployed on Kubernetes.
```

5. Scale to 4 replicas

```
ronak@ronak:~/devops-task-manager$ kubectl scale deployment task-manager-deployment --replicas=
4
deployment.apps/task-manager-deployment scaled
ronak@ronak:~/devops-task-manager$ kubectl get deployments
NAME                     READY   UP-TO-DATE   AVAILABLE   AGE
task-manager-deployment   4/4     4            4           2m
ronak@ronak:~/devops-task-manager$ kubectl get pods
task-manager-deployment-xxxxx-aaaaa       1/1     Running   0   2m
task-manager-deployment-xxxxx-bbbbb       1/1     Running   0   2m
task-manager-deployment-xxxxx-ccccc       1/1     Running   0   15s
task-manager-deployment-xxxxx-ddddd       1/1     Running   0   15s
```

6. Self-healing

```
ronak@ronak:~/devops-task-manager$ kubectl delete pod task-manager-deployment-xxxxx-aaaaa
pod "task-manager-deployment-xxxxx-aaaaa" deleted
ronak@ronak:~/devops-task-manager$ kubectl get pods
task-manager-deployment-xxxxx-bbbbb       1/1     Running             0   3m
task-manager-deployment-xxxxx-ccccc       1/1     Running             0   1m
task-manager-deployment-xxxxx-ddddd       1/1     Running             0   1m
task-manager-deployment-xxxxx-eeeee       1/1     ContainerCreating   0   2s
Kubernetes detected the missing Pod.
New Pod created automatically.
Desired Replicas: 4
Running Pods: 4
```

7. Pod logs and stop

```
ronak@ronak:~/devops-task-manager$ kubectl logs task-manager-deployment-xxxxx-bbbbb
Student Task Manager running on port 3000
ronak@ronak:~/devops-task-manager$ minikube stop
* Stopping node "minikube" ...
* 1 node stopped.
```

Conclusion

The application was deployed on a Minikube cluster with a NodePort service, scaled from 2 to 4 replicas, and Kubernetes self-healing was demonstrated.